

COURSE SYLLABUS

CMSC 56: Discrete Mathematical Structure in Computer Science I

I. COURSE OVERVIEW & DESCRIPTION

Course Title: Discrete Mathematical Structure in Computer Science I

Academic Unit: College of Science and Mathematics

Coverage: Principles of Logic, Set Theory, Relations and Functions, Boolean and Linear Algebra.

Mathematics is a broad field generally viewed as continuous or discrete. Continuous mathematics deals with values that vary without breaks, such as points on a real line. Discrete mathematics, on the other hand, deals with distinct and separate objects, such as integers, graphs, sets, and logical statements.

Why does this matter to a Computer Science student? Discrete mathematics is the fundamental foundation of Computer Science. The way computers represent information, make logical decisions, organize data structures, and execute problem-solving algorithms relies heavily on discrete structures and rigorous mathematical reasoning. This course builds the mathematical and logical thinking essential for your Computer Science journey.

II. COURSE GOALS & LEARNING OUTCOMES

Primary Goal: Master the fundamental principles of logic, set theory, relations, functions, Boolean algebra, and linear algebra applicable to computer systems.

Course Learning Outcomes (CLOs): Upon completion, students should be able to:

- **CLO 1:** Apply rigorous properties of sets, partitions, and Cartesian products to solve discrete problems.
- **CLO 2:** Evaluate binary relations on finite structures (e.g., strings) and identify equivalence or order properties.
- **CLO 3:** Analyze algorithmic processes using mathematical functions and quantify function complexity.
- **CLO 4:** Formalize syntax and semantics for both propositional and predicate logical systems.
- **CLO 5:** Translate complex natural language statements (English) into precise symbolic logical structures.
- **CLO 6:** Construct sound logical arguments, evaluate sentence validity, and execute accurate inference steps.
- **CLO 7:** Design, optimize, and minimize simple switching logic circuits for given Boolean functions.
- **CLO 8:** Apply core principles of linear algebra and matrix operations to computational problems.

III. DETAILED COURSE OUTLINE

MODULE I: SET THEORY, RELATIONS & FUNCTIONS

- **A. Sets:** Basic definitions, Venn diagrams, set operations.
- **B. Identities:** Set laws, algebraic proofs, membership tables.
- **C. Advanced Sets:** Partitions, Cartesian products, Power sets.
- **D. Relations:** Definition, representations, directed graphs.
- **E. Operations:** Composition, closures, inverse relations.
- **F. Properties:** Reflexive, symmetric, transitive, antisymmetric.
- **G. Orders:** Equivalence relations, Partial & Total Orders.

MODULE II: PRINCIPLES OF LOGIC

- **A. Propositions:** Compound statements, truth tables.
- **B. Equivalences:** Tautologies, contradictions, logical laws.
- **C. Conditionals:** Implication, converse, inverse, contrapositive.
- **D. Arguments:** Valid/invalid rules of inference, fallacies.
- **E. Predicates:** Quantifiers ($\forall$, $\exists$), domain, negation rules.
- **F. Predicate Arguments:** Universal instantiation/generalization.

MODULE III: BOOLEAN & LINEAR ALGEBRA

- **A. Boolean Algebra:** Axioms, duality, canonical forms.
- **B. Functions:** Sum-of-Products (SOP), Product-of-Sums (POS).
- **C. Logic Gates:** AND, OR, NOT, NAND, NOR, XOR, XNOR.
- **D. Minimization:** Algebraic simplification, circuit design.
- **E. Karnaugh Maps:** 2, 3, and 4-variable K-Map optimization.
- **F. Quine-McCluskey:** Tabular method, prime implicants.
- **G. Linear Algebra:** Vectors, dot products, CS applications.
- **H-I. Matrices:** Matrix operations, special matrices, system solution.

IV. GRADING SCHEME & EVALUATION

Course requirements are divided across three major module areas. Submissions are managed via Canvas LMS or in-person as directed. Email submissions are not accepted.

Assessment Component	Weight Allocation
Chapter Assessments / Long Examinations (End of Module Exams)	45%
Assignments / Exercises / Quizzes / Portfolio & Reflections	35%
Final Assessment / Project (Comprehensive)	15%
Attendance & Active Class Participation	5%
TOTAL	100%

EVALUATION SCALE (PASSING GRADE: 60%)

Percentage Range	Final Grade	Percentage Range	Final Grade
[96, 100]	1.00	[75, 80)	2.25
[92, 96)	1.25	[70, 75)	2.50
[88, 92)	1.50	[65, 70)	2.75
[84, 88)	1.75	[60, 65)	3.00
[80, 84)	2.00	Below 60	5.00

* Note: No grade of 4.0 is given in this course.

SOLUTION MARKING SCHEME RUBRIC

- **85% Points:** Correct answer with minor process/calculation errors.
- **75% Points:** Incorrect answer, but follows correct complete process logic.
- **50% Points:** Correct/incorrect answer but solution has major logical errors.
- **50% Points:** Correct final answer, but solution is incomplete/unsupported.
- **25% Points:** Completely incorrect process and answer.
- **0% Points:** No submission / zero effort shown.

V. COURSE POLICIES & ASSESSMENT GUIDELINES

- **Assignments & Portfolios:** Given periodically with 1–2 week submission windows. Late submissions are accepted only for meritorious cases with a maximum ceiling grade of 50% of the total score. Re-checking complaints are entertained within one week of work return.
- **Quizzes:** Administered during F2F classes or via Canvas LMS. Timed online quizzes require stable connectivity. *No special make-up quizzes* are administered for missed quizzes.
- **Chapter & Final Exams:** Long exams occur F2F. Missed exams require an approved Excuse Slip from the Office of the College Secretary (OCS). Special make-up exams are granted solely for highly meritorious, officially excused cases.
- **INC, DRP, and Failure (5.0):** Unsubmitted items 1 week after finals receive a score of zero. Grade of INC is given only for officially excused missed final assessments. Unofficial dropping or failing standing (<60%) results in 5.0 unless officially dropped via OCS.

VI. RECOMMENDED REFERENCES

1. Johnsonbaugh, R. (2001). *Discrete Mathematics* (5th ed.). Pearson Education Asia.
2. Rosen, K. (1999). *Discrete Mathematics and its Applications*. McGraw-Hill.
3. Dettman, J. (1974). *Introduction to Linear Algebra and Differential Equations*. Dover Publishing.
4. Golub, G. & Van Loan, C. (1989). *Matrix Computations* (2nd ed.). Johns Hopkins University Press.
5. Grimaldi, R. (1994). *Discrete and Combinatorial Mathematics: An Applied Introduction* (3rd ed.). Addison-Wesley.
6. Kolman, B. & Busby, R. (1987). *Discrete Mathematical Structures for Computer Science* (2nd ed.). Prentice-Hall.
7. Levy, L. (1980). *Discrete Structures of Computer Science*. John Wiley & Sons, Inc.
8. Manna, Z. & Waldinger, R. (1985). *The Logical Basis for Computer Programming: Deductive Reasoning*. Addison-Wesley.
9. Mott, J., et al. (1983). *Discrete Mathematics for Computer Scientists* (2nd ed.). Reston Publishing.

VII. HOUSE RULES, NETIQUETTE & ACADEMIC INTEGRITY

"Tell me and I forget, teach me and I may remember, involve me and I learn." — Benjamin Franklin

- **Netiquette & Collaboration:** Observe respectful online and offline communication. Practice peer discussion and solve extra practice problems continuously. *"The only way to learn mathematics is to do mathematics."* — Paul Halmos
- **Academic Integrity:** Uphold **Honor and Excellence**. Cheating or academic dishonesty will face strict disciplinary action under Sections 2 & 25 of the UP Faculty Manual.
- **Intellectual Property & Privacy:** Course materials are strictly for enrolled students. Do not distribute resources outside Canvas LMS without explicit written consent ([UP IP Guidelines](#)).

VIII. INSTITUTIONAL POLICY ON ARTIFICIAL INTELLIGENCE (AI) USE

Conditional AI Permissibility: Generative AI tools (e.g., Copilot, ChatGPT) may be used on designated assignments as explicitly stated in assignment prompts. When utilized, AI tools must be cited transparently. Uncited AI output is treated as academic misconduct. Students are solely responsible for verifying the accuracy of AI outputs and ensuring data privacy.

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